

ALMATY INTL, Kazakhstan

WMO: 368720

Lat: 43.352N

Lon: 77.041E

Elev: 681

StdP: 93.41

Time Zone: 6.00 (E06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-18.2	-24.0	0.5	-21.7	-20.5	0.7	-18.2	7.3	1.5	5.8	-0.2	1.3	190	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	35.1	17.7	33.8	17.5	32.0	17.2	19.9	29.5	19.0	29.3	18.4	28.5	3.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.2	22.7	15.9	12.3	22.4	14.9	11.5	21.9	60.0	29.8	57.2	29.0	54.8	28.2	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	5.8	4.9	DB	-24.4	37.4	4.3	1.3	-27.5	38.4	-30.1	39.2	-32.5	39.9	-35.6	40.9	(4)
(5)			WB	-24.6	21.8	4.2	1.7	-27.7	23.0	-30.2	24.0	-32.6	25.0	-35.7	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-6.7	-3.9	4.6	12.0	16.7	21.6	24.1	22.7	17.5	10.4	2.2	-4.1	(6)	
	DBStd	11.62	5.96	6.10	5.71	4.33	3.57	2.98	2.67	3.35	3.99	4.55	5.58	5.32	(7)	
	HDD10.0	1845	519	388	179	29	2	0	0	2	49	239	438		(8)	
	HDD18.3	3593	777	622	426	191	73	8	1	5	63	248	484	696	(9)	
	CDD10.0	1784	0	0	12	90	210	349	437	394	226	61	6	0	(10)	
	CDD18.3	489	0	0	0	2	23	107	178	141	36	2	0	0	(11)	
	CDH23.3	6067	0	0	2	62	321	1194	2158	1741	542	46	1	0	(12)	
	CDH26.7	2616	0	0	0	10	73	463	1054	816	189	10	0	0	(13)	
(14) Wind	WSAvg	2.2	1.7	1.9	2.3	2.5	2.5	2.5	2.5	2.4	2.2	1.8	1.8	1.8	(14)	
(15) Precipitation	PrecAvg	786	35	38	69	113	124	88	84	48	34	54	56	43	(15)	
	PrecMax	1035	56	52	127	197	200	140	203	107	64	121	137	85	(16)	
	PrecMin	487	14	20	33	18	33	37	31	2	7	15	22	19	(17)	
	PrecStd	140	12	9	23	52	43	31	43	23	17	32	31	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	11.5	22.2	28.2	31.1	34.9	37.2	36.8	33.1	28.0	20.4	10.1	(19)	
		MCWB	3.2	6.2	11.4	14.6	17.0	18.0	18.0	17.7	15.6	13.2	10.9	5.6	(20)	
	2%	DB	4.1	8.0	19.1	25.1	28.8	32.8	35.4	35.0	31.0	24.0	16.1	6.5	(21)	
		MCWB	1.6	4.4	10.2	13.8	16.5	17.6	17.9	17.4	15.7	12.4	9.6	3.5	(22)	
	5%	DB	2.2	5.8	16.1	22.9	26.9	31.1	34.0	33.2	29.0	21.1	12.8	4.1	(23)	
		MCWB	0.3	2.9	9.2	12.8	15.6	17.3	17.9	17.1	14.6	11.4	7.6	2.1	(24)	
	10%	DB	0.9	3.2	13.8	20.8	25.0	29.5	32.1	31.2	26.8	18.9	10.0	2.3	(25)	
		MCWB	-0.5	1.2	8.3	12.3	14.9	17.1	17.6	16.5	14.0	10.4	6.2	0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	6.8	12.8	17.1	19.2	20.6	21.9	20.4	17.9	14.4	12.4	5.7	(27)	
		MCDB	5.9	10.2	19.9	24.6	27.5	29.3	31.2	30.3	29.0	23.2	18.9	7.8	(28)	
	2%	WB	1.9	4.5	11.1	15.2	17.8	19.5	20.1	18.9	16.2	13.2	10.0	3.8	(29)	
		MCDB	3.6	7.4	17.6	21.8	25.8	28.8	30.0	30.1	27.1	21.7	15.2	6.0	(30)	
	5%	WB	0.6	2.9	9.6	13.9	16.7	18.6	19.2	18.2	15.4	12.2	8.0	2.2	(31)	
		MCDB	1.8	5.4	15.1	20.3	24.5	28.1	29.6	29.2	26.3	18.8	11.5	3.9	(32)	
	10%	WB	-0.6	1.5	8.4	12.9	15.6	17.9	18.5	17.5	14.5	11.2	6.4	0.8	(33)	
		MCDB	0.5	3.0	13.4	19.4	23.0	26.9	29.0	28.2	24.9	17.4	9.5	2.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.0	9.3	11.4	12.3	13.2	13.9	14.8	15.1	14.8	13.2	10.0	8.8	(35)	
		MCDBR	11.6	11.8	16.1	16.8	16.1	16.4	17.0	17.1	17.9	17.1	14.6	11.4	(36)	
	5% WB	MCWBR	8.9	8.4	8.7	8.1	7.4	6.4	5.8	5.9	6.6	7.5	8.5	8.4	(37)	
		MCWBR	10.2	10.8	14.3	13.7	14.4	14.3	14.9	15.4	16.1	14.3	13.1	10.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.380	0.441	0.478	0.452	0.453	0.446	0.417	0.400	0.406	0.373	0.328	(40)	
		taud	2.147	2.038	1.940	1.911	2.056	2.111	2.155	2.225	2.208	2.086	2.102	2.160	(41)	
	Ebn at Noon		763	784	782	789	825	824	825	838	820	746	702	728	(42)	
		Edh at Noon	99	131	165	183	163	155	147	132	125	124	103	89	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.01	2.92	3.82	4.93	6.21	6.60	6.52	6.09	4.89	3.19	1.95	1.56	(44)	
	RadStd		0.15	0.23	0.30	0.42	0.42	0.27	0.39	0.30	0.26	0.28	0.19	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-203	-252	+139	+72	

Nomenclature: See separate page